

Rasch-built Overall Disability Scale (R-ODS) for immune-mediated peripheral neuropathies

S.I. van Nes, MD
E.K. Vanhoutte, MD
P.A. van Doorn, MD,
PhD
M. Hermans, MD
M. Bakkers, MD
K. Kuitwaard, MD
C.G. Faber, MD, PhD
I.S.J. Merkies, MD,
PhD

Address correspondence and
reprint requests to Dr. S.I. van
Nes, Erasmus MC, University
Medical Center Rotterdam,
Department of Neurology, PO
Box 2040, 3000 CA, Rotterdam,
the Netherlands
s.vannes@erasmusmc.nl

ABSTRACT

Objective: To develop a patient-based, linearly weighted scale that captures activity and social participation limitations in patients with Guillain-Barré syndrome (GBS), chronic inflammatory demyelinating polyradiculoneuropathy (CIDP), and gammopathy-related polyneuropathy (MGUSP).

Methods: A preliminary Rasch-built Overall Disability Scale (R-ODS) containing 146 activity and participation items was constructed, based on the WHO International Classification of Functioning, Disability and Health, literature search, and patient interviews. The preliminary R-ODS was assessed twice (interval: 2–4 weeks; test-retest reliability studies) in 294 patients who experienced GBS in the past ($n = 174$) or currently have stable CIDP ($n = 80$) or MGUSP ($n = 40$). Data were analyzed using the Rasch unidimensional measurement model (RUMM2020).

Results: The preliminary R-ODS did not meet the Rasch model expectations. Based on disordered thresholds, misfit statistics, item bias, and local dependency, items were systematically removed to improve the model fit, regularly controlling the class intervals and model statistics. Finally, we succeeded in constructing a 24-item scale that fulfilled all Rasch requirements. “Reading a newspaper/book” and “eating” were the 2 easiest items; “standing for hours” and “running” were the most difficult ones. Good validity and reliability were obtained.

Conclusion: The R-ODS is a linearly weighted scale that specifically captures activity and social participation limitations in patients with GBS, CIDP, and MGUSP. Compared to the Overall Disability Sum Score, the R-ODS represents a wider range of item difficulties, thereby better targeting patients with different ability levels. If responsive, the R-ODS will be valuable for future clinical trials and follow-up studies in these conditions. *Neurology*® 2011;76:337–345

GLOSSARY

ALDS = AMC Linear Disability Score; **CI** = confidence interval; **CIDP** = chronic inflammatory demyelinating polyradiculoneuropathy; **CTT** = classic test theory; **DIF** = differential item functioning; **GBS** = Guillain-Barré syndrome; **ICF** = International Classification of Functioning, Disability and Health; **IRT** = item response theory; **MGUSP** = gammopathy-related polyneuropathy; **MMN** = multifocal motor neuropathy; **ODSS** = Overall Disability Sum Score; **PSI** = Person Separation Index; **R-ODS** = Rasch-built Overall Disability Scale; **RUMM2020** = Rasch unidimensional measurement model; **VSN** = Vereniging Spierziekten Nederland.

Disability has been proposed as the preferential level for measuring therapeutic response in immune-mediated neuropathies.¹ However, most disability scales used in these disorders are based on the classic test theory (CTT).^{2–7} A summary of the disadvantages of CTT has been provided.² Generally, patients are requested to complete all items of CTT-based scales, even though some may be irrelevant for their level of ability. Furthermore, a sum of item scores is often calculated assuming equal relevance and hence weighting of each item which is highly unlikely.^{2,8} Given these limitations, a modern approach like the Rasch technology is needed to develop a scale measuring disability (i.e., activity and social participation limitations) in patients with immune-mediated neuropathies.^{9,10} Rasch models the probability that a person will be able to complete an item, only dependent on the item difficulty and the person’s level of ability.^{9,11,12} It enables scales

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From the Department of Neurology (S.I.v.N., P.A.v.D., K.K., I.S.J.M.), Erasmus Medical Center Rotterdam, Rotterdam; Department of Neurology (E.K.V., M.H., M.B., C.G.F., I.S.J.M.), Maastricht University Medical Center, Maastricht; and Department of Neurology (I.S.J.M.), Spaarne Hospital, Hoofddorp, the Netherlands.

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measuring the same health construct to be equated on the same linear ruler optimizing comparisons of studies, meta-analyses, and systematic reviews.¹⁰

The AMC Linear Disability Score (ALDS) item bank and ACTIVLIM are modern disability scales.¹³⁻¹⁵ However, their calibration may not be representative to patients with immune-mediated neuropathies since these scales are not disease-specific. Therefore, we developed a linearly weighted scale at the activity and participation level, the Rasch-built Overall Disability Scale (R-ODS), specifically for patients with Guillain-Barré syndrome (GBS), chronic inflammatory demyelinating polyradiculoneuropathy (CIDP), and gammopathy-related polyneuropathy (MGUSP), and evaluated its validity and reliability aspects.

METHODS Participants. A total of 852 members of the Dutch society of neuromuscular disorders (Vereniging Spierziekten Nederland [VSN]) registered as having a polyneuropathy were initially requested to participate. For this study, we used data from patients who have been diagnosed with GBS, CIDP, or MGUSP (254 of the 511 members who returned their questionnaires). Additionally, 40 patients (11 GBS, 22 CIDP, and 7 MGUSP) were recruited at the university outpatient clinics of Rotterdam and Maastricht. All participants were recruited between January 2007 and July 2009. Participants aged 18 years and older, with a stable clinical condition, defined as an unchanged physical and functional condition over 2 months prior to the study as declared by the participant, were included.

Standard protocol approvals, registrations, and patient consents. The local Medical Ethics Committee approved the protocol. Written informed consent was obtained from all participants.

Questionnaire development. Step 1. International development procedures were applied to construct the preliminary R-ODS.^{12,16,17} Since we aimed to develop an outcome measure at the activity and participation level, we critically reviewed all potential items listed under these headings using the International Classification of Functioning, Disability and Health (ICF).¹⁸ The chapters learning and applying knowledge, general tasks and demands, and communication were excluded. We selected defined items from this ICF list with probable relevance to polyneuropathy patients based on known clinical characteristics. Efforts were made to describe these items in a concise, simple, and unambiguous way.

Step 2. A systematic Medline and Embase search was performed for literature published over the last 4 decades, reviewing all scales at the activity and participation levels in patients with any form of (poly)neuropathy. Reports published in English were identified with reference tracing using the key words GBS, CIDP, acquired/idiopathic (poly)(radiculo)neuropathy, polyn neuritis, gammopathy, dysimmune, paraprotein(a)emia, MGUSP, disability (scale), activity limitation, and handicap. Various handbooks of neurologic rating scales were also evaluated. Additionally, items were selected from an earlier semistruc-

tured interview focusing on daily activities addressed by patients with inflammatory polyneuropathies.⁵ Eventually, we succeeded in constructing the preliminary R-ODS containing 146 items with 3 response options: (0) “impossible to perform,” (1) “performed with difficulty,” and (2) “easily performed.” If an item was not applicable, the patient was requested to answer this item with the option (3) “not applicable.”

Step 3. The selected items were judged by 4 neuromuscular experts. Based on the clinical characteristics of GBS, CIDP, and MGUSP, items with insufficient face and content validity were removed.¹⁶

Procedures. Patients received the preliminary R-ODS plus the Overall Disability Sum Score (ODSS) with instructions by mail.⁶ They were requested to answer all questions by themselves and to answer “impossible to perform” when unable to complete an item or “able to perform, but with difficulty” when special devices or other forms of assistance were needed. After a period of 2–4 weeks, all participants were again requested to answer all questions of the preliminary R-ODS (test-retest reliability studies).

Rasch analyses. In the model construction, items scored as (3) “not applicable” were interpreted as missing data. Items with >10% missing values and patients with >10% unanswered items were omitted as a quality control procedure. Thereafter, the remaining responses of the preliminary R-ODS were analyzed using Rasch unidimensional measurement methods (RUMM2020), which is based on a mathematical model proposed by Rasch, a Danish mathematician.¹⁹ The finally constructed scale should be unidimensional, free from item bias, and without disordered thresholds or local dependency.^{10,20} To enhance the reading, we added a more detailed description of the Rasch model as appendix e-1 on the *Neurology*® Web site at www.neurology.org. We checked for possible item bias using the following person factors, categorized arbitrarily as follows: age (<60 years vs ≥60 years), gender (men vs women), diagnosis (GBS vs CIDP vs MGUSP), diagnosis category (acute [GBS] vs chronic [CIDP and MGUSP]), and duration of complaints (<2 years vs 2–5 years vs ≥5 years). Throughout the analyses, we continuously monitored the distribution of persons within the class intervals, the fit statistics, and the independent tests for unidimensionality of the scale. Items and persons not fulfilling Rasch model criteria were evaluated and removed one by one if needed.

A sample size of approximately 250 is needed to obtain a 99% confidence with a stable item calibration within ±0.5 logits, hence providing a stable model.²¹ We expected an acceptable degree of precision of the Rasch analyses, since a sample of 294 patients was available.

Validity. Four neuromuscular experts stated that the items of the final R-ODS were ordered as expected. The external construct validity was assessed by correlation with the ODSS (intra-class correlation coefficient after applying quantile regression analysis). To compare the difficulty range of the items of the ODSS and the final R-ODS in relation to the ability range of the patients in this population, we used the anchor-based approach in RUMM2020 (targeting validity study). Two identical items in the ODSS and the R-ODS (“turning a key in a lock” and “dressing upper body”) were used as anchors to place both measures and all patients on the same linear ruler.

Reliability. Internal reliability was examined by determining the Person Separation Index (PSI). In general, a PSI above 0.7 is considered as acceptable, indicating the ability to identify at least 2 groups of patients.²² Moreover, test-retest reliability studies

were performed to investigate whether hierarchy of item difficulty and patient ability location were consistent over time.²³ Reliability was quantified by calculation of the intraclass correlation coefficient using a one-way random effects analysis of variance model for group comparison.

Statistics and software. Rasch analyses were performed with the partial credit model as default, using RUMM2020 software. Further analyses were undertaken using Stata 10.0 for Windows XP. The *p* value was adjusted throughout the analyses, based on Bonferroni multiple testing corrections.²⁴

RESULTS General description of patients. A total of 511 of the 852 approached members of the Dutch society of neuromuscular disorders returned the first questionnaires (response rate of 60%). For the current study, we only used data of patients with a clinically stable form of immune-mediated polyneuropathy (174 GBS, 80 CIDP, and 40 MGUSP). Most patients (74%) reported walking difficulties; 47% mentioned at least moderate symptoms in their upper limbs. A total of 255 patients completed the second preliminary R-ODS assessment (87%). The basic characteristics of all participants are presented in table 1.

Data quality control (step 0). Based on experts' opinion, *n* = 12 items with insufficient face and content validity were removed (questionnaire development step 3). Also, another *n* = 36 items

with >10% missing values and *n* = 14 patients with >10% unanswered items were omitted as a quality control procedure.

Initial Rasch analysis on the preliminary R-ODS. The first Rasch analysis was performed on the remaining 98 items. The mean items' fit residual showed reasonable model fit (mean −0.188, SD 1.119). However, the mean persons fit residual was −0.363 with a SD exceeding the expected value of 1 (SD 1.428). The significant χ^2 probability implied no invariance of items. Initially, a proportion of 0.20 (95% confidence interval [CI] 0.18–0.23) of the *t* tests performed fell outside the ± 1.96 range, indicating multidimensionality.

Stepwise approach to fit data to the Rasch model. Step 1. Disordered thresholds were seen in 2 items ("driving a car" and "riding a bike"). Apparently, the response categories were conflicting for the patients. These 2 items were removed.

Step 2. Individual item and person fit statistics were inspected. Individual item fit statistics of 19 items demonstrated misfit to the model (fit residuals exceeding ± 2.5 or a significant χ^2 probability or both). These 19 items were removed one by one, starting with the item with the highest deviation. In addition, 2 patients showed misfit (fit residuals exceeding ± 2.5) and were also removed.

Step 3. In 6 items (e.g., "standing up from squat position," "shaving/epilating," and "vacuum cleaning"), men and women with equal ability levels responded in a significantly different manner (demonstrating uniform differential item functioning [DIF] on gender). Another 3 items demonstrated item bias on person factor diagnosis (category) (2 items had uniform DIF, 1 item nonuniform DIF). Finally, one item demonstrated item bias (uniform DIF) on the duration of complaints. All these items were removed one by one.

Step 4. A systematic evaluation of the correlation matrix findings (starting with the highest correlations ≥ 0.7 , then ≥ 0.6 , through to ≥ 0.28) was performed to identify local dependency (e.g., a person answering positively to the item "able to walk 1 km" will most likely also respond positively on the item "able to walk 100 m"). Item characteristic curves of each correlating item set were subsequently inspected to select the items that best fit the expected model curve. The item showing the most overdiscrimination or underdiscrimination was then removed. Eventually, a total of 43 items were stepwise removed.

Final R-ODS. After completing these procedures, we succeeded in constructing a 24-item scale which met the Rasch model expectations, the final R-ODS. Mean fit residuals for items were −0.238 (SD 0.853)

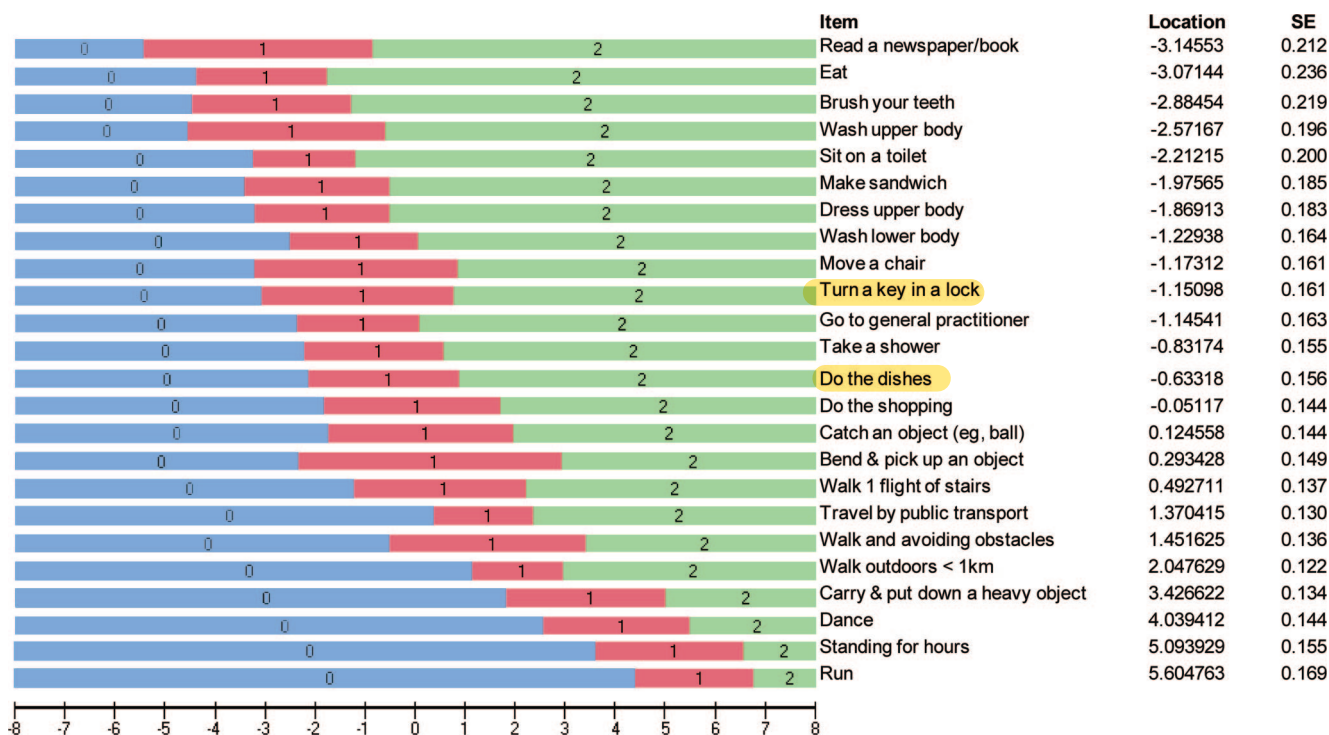
Table 1 Basic characteristics of study population (first sample, *n* = 294)

Diagnosis, <i>n</i> (%)	Values
GBS	174 (59)
CIDP	80 (27)
MGUSP	40 (14)
Age, <i>y</i> , mean (SD), range	60.2 (13.5), 18–91
Gender, <i>n</i> (%)	
Women	132 (45)
Men	162 (55)
Duration of symptoms, <i>y</i> , mean (SD), range	8.2 (7.8), 0.5–40
ODSS arm grade, <i>n</i> (%) ^a	
≤1 (normal or minor symptoms, not affecting any)	123 (53.2)
2 (moderate symptoms, not preventing any)	87 (37.7)
3 (severe symptoms, preventing at least one not all)	20 (8.7)
≥4 (severe symptoms, preventing all or no purposeful movements)	1 (0.4)
ODSS leg grade, <i>n</i> (%) ^a	
0 (normal)	59 (25.5)
1–2 (walking affected, with or without gait disturbance)	103 (44.6)
3 (unilateral support)	40 (17.3)
4 (bilateral support)	10 (4.3)
≥5 (wheelchair)	19 (8.2)

Abbreviations: CIDP = chronic inflammatory demyelinating polyradiculoneuropathy; GBS = Guillain-Barré syndrome; MGUSP = gammopathy-related polyneuropathy; ODSS = Overall Disability Sum Score.

^a Corresponding ODSS grades were only collected in 231 of the 294 patients.

Figure 1 Threshold map and overall item difficulty locations for each item of the final Rasch-built Overall Disability Scale (R-ODS)



Blue sections (0) = impossible to perform; red sections (1) = performed with difficulty; green sections (2) = easily performed. The length of the colored sections at the left panel represents the number of patients giving a certain response. A gradual shift is seen: the blue section of the item "reading a newspaper/book" is much shorter than the blue section of the item "running." Patients considered "reading a newspaper/book" the easiest item and "running" the most difficult item. The right panel presents the overall item difficulty locations for each item.

and for persons 0.352 (SD 0.839). The χ^2 probability was $p = 0.29$, thereby supporting the presence of invariance of item difficulty across the scale. Based on the first principal components analysis, 2 subsets of items were formed, with the 6 most positively loaded vs the 6 most negatively loaded items. The independent t tests between these 2 subsets were significant at the 5% level in 14 of the 261 patients (0.053%, CI 0.027%–0.080%), which indicates acceptable unidimensionality.

In the final R-ODS scale, the item "reading a newspaper/book" was the easiest to perform while "running" turned out to be the most difficult task (figure 1). Item difficulty ranges from -3.15 to 5.60 logits and patient ability level from -5.35 to 8.11 logits. Table 2 provides a nomogram allowing the translation of raw sumscores of the final R-ODS (range 0–48) to logits, placing patients' estimates on the same log-odds units (logit) scale. Since logits are difficult to interpret instinctively, we have converted the person locations into a centile metric score with values ranging from 0 (most severe activity and social participation limitations) to 100 (no activity and social participation limitations).

Validity. The intraclass correlation coefficient between the final R-ODS and the ODSS was 0.85,

indicating good external construct validity. Furthermore, the population of patients examined demonstrated a ceiling effect of 5.8% on the final R-ODS vs 19.4% ceiling effect on the ordinal ODSS (Student t test $p < 0.0001$). As shown in figure 2, item difficulties of the ODSS ranged from -1.869 to 1.336 logits (span of 3.205 logits). By contrast, the final R-ODS demonstrated a wider range of item difficulties (span of 8.750 logits).

Reliability. Internal reliability remained good, indicated by a PSI of 0.97 for the final R-ODS. Also, test-retest reliability was good: item hierarchy and patients' location were located within the 95% CI lines, reflecting good invariance (figure 3, A and B).

DISCUSSION We developed the R-ODS, a linearly weighted outcome measure that specifically captures activity and social participation limitations in patients with GBS, CIDP, and MGUSP. This questionnaire was constructed using patient's perception of their ability to perform daily and social activities. All 24 items of the final R-ODS met the Rasch model expectations. Although the clinical characteristics of clinically stable GBS, CIDP, and MGUSP patients are slightly different, their response pattern

Table 2 Nomogram ^a		
R-ODS summed raw score	Rasch person location (logits)	Centile metric
0	−6.95	0
1	−6.03	6
2	−5.36	11
3	−4.87	14
4	−4.48	16
5	−4.14	19
6	−3.84	21
7	−3.57	22
8	−3.32	24
9	−3.09	26
10	−2.87	27
11	−2.66	28
12	−2.46	30
13	−2.26	31
14	−2.07	32
15	−1.88	34
16	−1.70	35
17	−1.52	36
18	−1.33	37
19	−1.15	39
20	−0.97	40
21	−0.79	41
22	−0.61	42
23	−0.42	43
24	−0.24	45
25	−0.05	46
26	0.14	47
27	0.34	48
28	0.53	50
29	0.73	51
30	0.94	52
31	1.15	54
32	1.36	55
33	1.58	57
34	1.81	58
35	2.04	60
36	2.28	61
37	2.54	63
38	2.80	65
39	3.09	67
40	3.40	69
41	3.74	71
42	4.11	73
43	4.54	76
44	5.03	80

—Continued

Table 2 Continued		
R-ODS summed raw score	Rasch person location (logits)	Centile metric
45	5.59	83
46	6.25	88
47	7.07	93
48	8.11	100

Abbreviations: R-ODS = Rasch-built Overall Disability Scale; RUMM2020 = Rasch unidimensional measurement model.

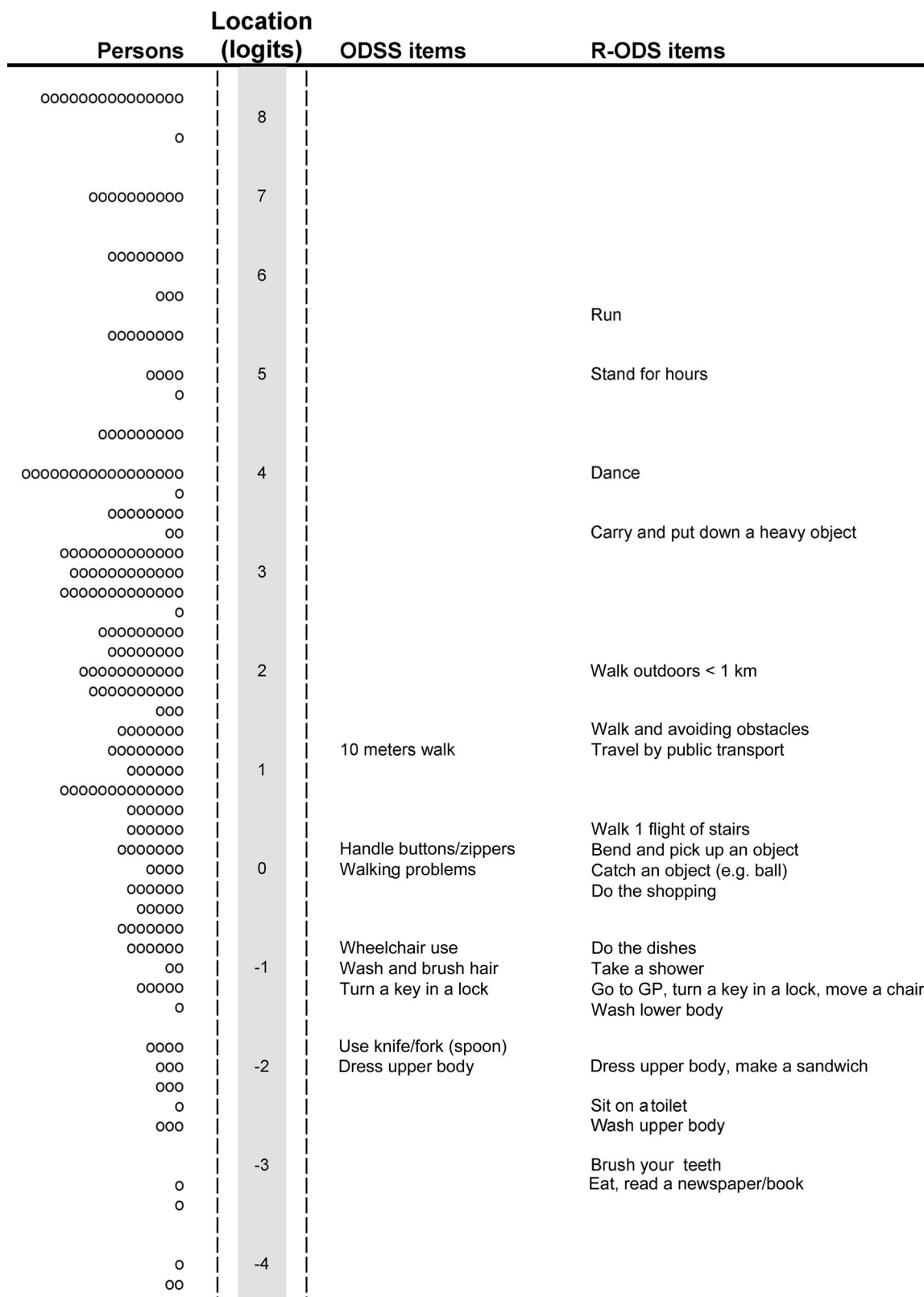
^a This nomogram allows the translation of raw sumscores of the final R-ODS (range 0–48) to logits or to a centile metric score with values ranging from 0 (most severe activity and social participation restrictions) to 100 (no activity and social participation limitations). The corresponding logits in relation to the raw summed scores is provided by the RUMM software. For convenience, the model also provides an equation for rescaling the logits to more digestible numbers (centile metric).

was similar in this study. Only 3 items of the preliminary scale were removed due to item bias for the person factor diagnosis.

Since calibration of the R-ODS is based on the characteristics of our study population, we also questioned whether this population represents patients included in clinical trials. The baseline characteristics of patients in trials often report a sumscore of the ordinal ODSS of around 4. Moreover, inclusion criteria sometimes include minimum sumscores of the ODSS (e.g., at least 2 in the Randomized Methotrexate CIDP trial, evaluating the efficacy of methotrexate in patients with CIDP requiring immunoglobulin or corticosteroids) or a minimum ODSS arm or leg grade (e.g., at least arm grade 2 or leg grade 1 in a trial evaluating the efficacy of IV immunoglobulin in MGUSP).^{25–27} The basic characteristics of our sample show arm grades ≥ 2 in 47% and leg grades ≥ 1 in 74% of the patients. This implies sufficient disability in the sample to calibrate the R-ODS for future use in CIDP and MGUSP trials. Only 30% of our sample was unable to walk independently (table 1, ODSS leg grade ≥ 3), corresponding to a GBS disability score of ≥ 3 , which is often used as criterion to enroll patients in GBS trials.^{28,29} Clearly, patients with GBS in the acute phase may often have more severe symptoms, signs, and functional deficit than our clinically stable patients who experienced GBS in the past. It is therefore conceivable that the weights of the items might have been different if more severely affected patients were included. However, since the PSI of the R-ODS turned out to be extremely high (0.97), it is reasonable to assume that this scale will also be able to differentiate between the various functional phase changes in patients with GBS.³⁰

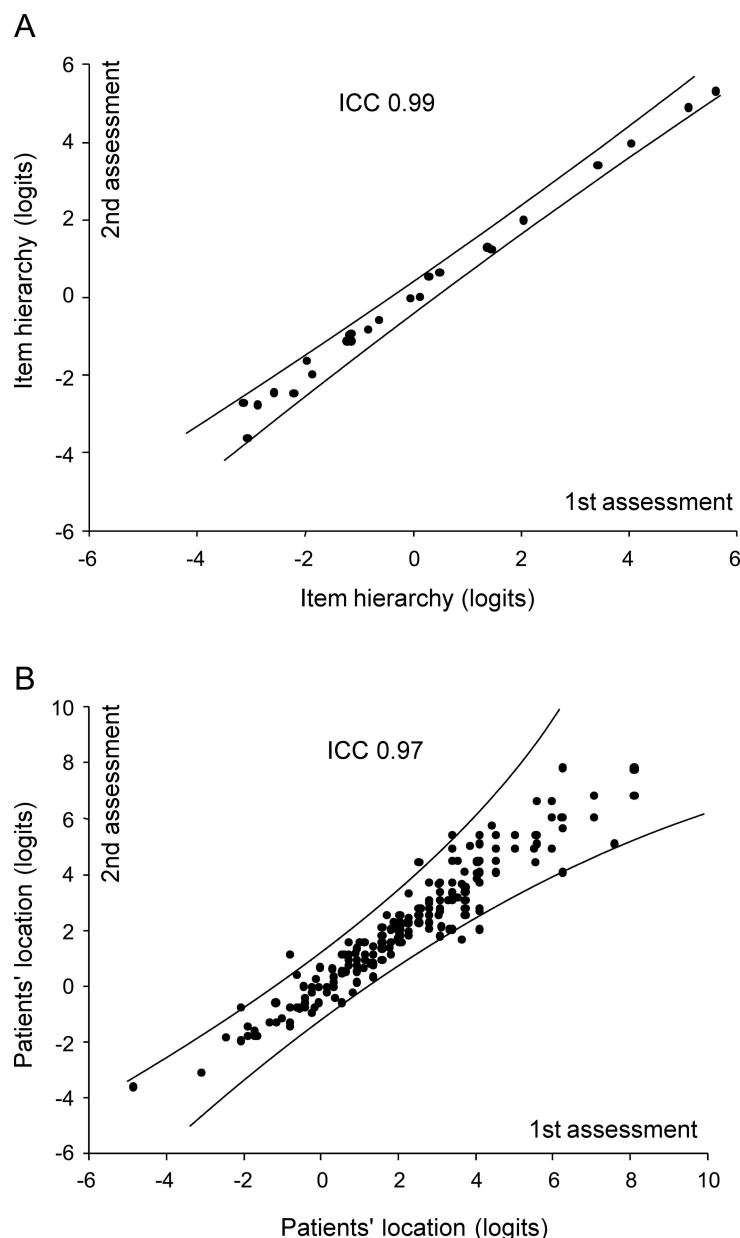
The ALDS item bank and ACTIVLIM are both linearly weighted scales quantifying functional status.

Figure 2 Person locations vs item locations



The numbers 8 to -6 under "Location (logits)" represent corresponding logit scale of a ruler. The open dots under "Persons" represent the number of patients having a person location (ability level) at this point on the ruler. Each open dot corresponds to the location of one patient on the ruler. At the right side of the ruler, item locations (item difficulty) of the Overall Disability Sum Score (ODSS) and the final Rasch-built Overall Disability Scale (R-ODS) are presented, enabling comparison through targeting (e.g., item difficulty range in relation to the different levels of ability in the sample). See also appendix e-1, explaining the findings of this target plot step by step; see section: "Understanding the provided centile metric for the R-ODS," in particular figures e-10A through e-10E. GP = general practitioner.

Figure 3 Test-retest reliability of the final R-ODS



(A) Item difficulty hierarchy of the final R-ODS in the first vs the second assessment. All items (dots) were located within the 95% confidence interval (CI) (solid lines) reflecting ideal reliability. (B) Patients' location in the first assessment vs the second assessment. Almost all persons (dots) were located within the 95% CI (solid lines). ICC = intraclass correlation coefficient.

The ALDS focuses on chronic conditions and ACTIVLIM on various neuromuscular disorders. We compared the item locations of the R-ODS with the locations of similar items of the ALDS and the ACTIVLIM.^{13,15} Not only do the person locations (logits) differ considerably between these scales and the R-ODS, but also the item difficulty ordering differs noticeably. This strengthens the idea that disease-specific scales should be constructed. In our view, although activity and social participation limitations are considered not being disease-specific, the significance and weight of corresponding items may

appear specific for the various illnesses. Support for this implication is also obtained when looking at the responses of patients with multifocal motor neuropathy (MMN).³¹ Our primary aim was to construct an outcome measure suitable for use in future trials evaluating therapeutic efficacy in a wide range of immune-mediated neuropathies. Therefore, at first not only the current patient population (GBS, CIDP, and MGUSP), but also patients with stable MMN were included. However, the primary analyses showed the patients with MMN behaving strongly differently. Fine motor upper limb items like “turn a key in a lock” and “make a sandwich” were considered much more difficult to accomplish by the MMN group (thus showing significant uniform DIF on person factor “diagnosis”). Their responses to the above items resembled the difficulty levels of items like “walking uphill” and “standing for hours” in our patients with GBS, CIDP, or MGUSP. We decided to not split the deviating items for the various illnesses. Instead, we are currently constructing a MMN-specific activity and participation scale to bypass these limitations.

Both item response theory (IRT) and Rasch measurement estimate the probability that a person with a given ability level will be able to complete an item/task.^{9,11,32} The ALDS is based on IRT, which aims to find the best model that explains the data, whereas Rasch observed rating scale data have to satisfy the model expectations.³³⁻³⁵ As a result, in IRT often a model with generally fewer restrictions is chosen to explain all data. We developed the R-ODS using the Rasch model since it is considered a strong model with many restrictions aiming to satisfy mathematical requirements necessary to achieve fundamental measurement.³⁶

There are some limitations that should be addressed. Most participants (86%) were members of a patient organization (VSN). We could not verify their diagnosis, but the board of the VSN has ascertained us that all patients came from centers with great expertise in neuromuscular disorders, ensuring the right diagnosis. Written instructions were given how to respond when assistance or special devices were needed. Nevertheless, there might be some bias of the responses given, depending on the daily and social situation of patients and adaptations made. Furthermore, due to cultural and geographic differences, items may be applicable to the Netherlands but not necessarily elsewhere. The ability of the R-ODS to detect relevant clinical changes over time (responsiveness) also needs further evaluation. Its responsiveness and international applicability is currently being investigated in patients with newly diagnosed immune-mediated neuropathies as part of

the international multicenter Peripheral Neuropathy Outcome Measures Standardization study.

Nevertheless, the R-ODS seems to be a valid and reliable outcome measure capturing activity and social participation limitations in patients with GBS, CIDP, and MGUSP. Compared to the ODSS, the R-ODS represents a wider range of item difficulties, thereby showing a better targeting of the different ability levels of these patients. If its responsiveness can be demonstrated, we expect that the R-ODS will be valuable for future clinical trials and follow-up studies in patients with these disorders.

AUTHOR CONTRIBUTIONS

Study design: C.G.F., I.S.J.M. Recruitment, data collection: S.I.v.N., E.K.V., P.A.v.D., M.H., M.B., K.K., C.G.F., I.S.J.M. Statistical analysis: C.G.F., I.S.J.M. Final report: all.

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DISCLOSURE

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